

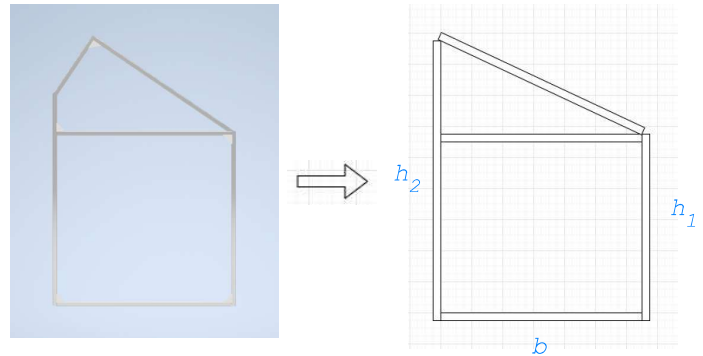
Sample Calculations:

1. Heat loss:

$h_1 := 820 \text{ mm}$ $h_2 := 1000 \text{ mm}$ height of back panel and front panel
 $l := 1000 \text{ mm}$ $b := 850 \text{ mm}$ length and base of greenhouse foundation
 $t := 0,003 \text{ m}$ thickness of panels

The shape of the side profile of the greenhouse has been approximated to simplify the calculations.

Assume ground surface area to not culminate any heat loss, but rather supply thermal mass to stabilize heat differences.



$$A := l \cdot h_1 \cdot 2 + l \cdot h_2 + \left(2 \cdot b \cdot \frac{(h_2 - h_1)}{2} + 2 \cdot b \cdot h_1 \right) \quad \text{area of all sides of perspex}$$

$$A = 4,187 \text{ m}^2$$

$$U := 5,7 \frac{\text{W}}{\text{m}^2 \text{ K}} \quad \text{for 3mm thick perspex from Maizey plastics datasheet}$$

$$\Delta T := 5 \text{ K} \quad \text{for a temperature difference of } 5^\circ\text{C}$$

$$Q := A \cdot U \cdot \Delta T \quad \text{Heat Loss} \quad (1.1)$$

$$Q = 119,3295 \text{ W}$$

$$\text{Daily} := Q \cdot 4 \text{ hr} \quad \text{assuming 4 hours of peak cold temperature difference of } 5^\circ\text{C during the night}$$

$$\text{Daily} = 0,4773 \text{ kW hr}$$

$$\Delta T_{\text{coldest}} := (15 - 0) \text{ K} \quad \text{assuming outside temperatures can reach } 0^\circ\text{C at night and minimum setpoint temperature is } 15^\circ\text{C}$$

$$Q_{\text{coldest}} := A \cdot U \cdot \Delta T_{\text{coldest}} \quad (1.2)$$

$$Q_{\text{coldest}} = 357,9885 \text{ W}$$

2. Vent Actuation:

$$\rho_{\text{perspex}} := 1,19 \frac{\text{g}}{\text{cm}^3} \quad \text{density of perspex}$$

$$w_{\text{vent}} := 270 \text{ mm} \quad \text{width of vent}$$

$$m_{\text{vent}} := l \cdot w_{\text{vent}} \cdot t \cdot \rho_{\text{perspex}} = 0,9639 \text{ kg} \quad \text{mass of vent}$$

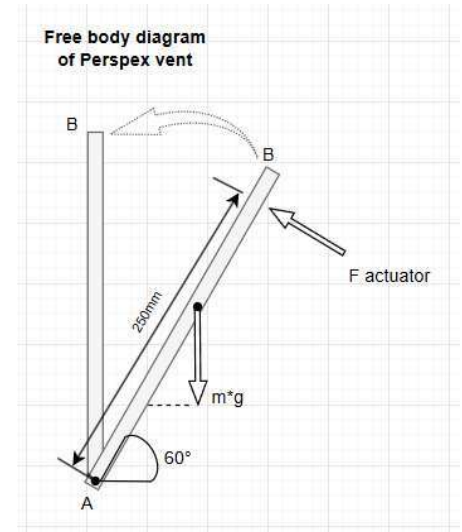
$$\theta := 60^\circ \quad \text{angle of inclination of vent}$$

$$F_{\text{weight}} := m_{\text{vent}} \cdot 9,81 \frac{\text{m}}{\text{s}^2} = 9,4559 \text{ N} \quad \text{weight of vent acting at centroid}$$

$$\sum M_A := 0$$

$$M_{\text{weight}} := \left(-F_{\text{weight}} \right) \cdot \frac{w_{\text{vent}}}{2} \cdot \cos(\theta) = 1,2158 \text{ N m} \quad \text{moment around point A due to weight}$$

$$F_{\text{actuator}} := \frac{M_{\text{weight}}}{0,25 \text{ m}} = 4,8632 \text{ N} \quad \text{assuming that the linear actuator is connected to the panel at 250mm from the hinges at A and that it is initially acting perpendicularly to the panel}$$



3. Cooling Power:

$$\rho := 1,2 \frac{\text{kg}}{\text{m}^3} \quad \text{Air density} \quad \text{a 50 CFM small intake fan approximates to } 0.02359 \text{ m}^3/\text{s}$$

$$C := 0,6 \quad \text{Discharge coefficient}$$

$$A_{\text{vent}} := 0,27 \text{ m}^2 \quad \text{Area of vent}$$

$$g := 9,81 \frac{\text{m}}{\text{s}^2}$$

$$H := 1,2 \text{ m} \quad \text{Height difference between intake and vent}$$

$$T := 293 \text{ K} \quad \text{Average absolute Temperature assumed to be } 20^\circ\text{C}$$

$$V_{\text{vent}} := C \cdot A_{\text{vent}} \cdot \sqrt{2 \cdot g \cdot H \cdot \frac{\Delta T}{T}} \quad \text{Bouyancy driven ventilation estimation using stack effect formula} \quad (3.1)$$

$$V_{\text{vent}} = 0,1027 \frac{\text{m}^3}{\text{s}}$$

$$V_{\text{fans}} := 0,05 \frac{\text{m}^3}{\text{s}} \quad \text{Volumetric flowrate of two intake fans}$$

$$c_p := 1005 \frac{\text{J}}{\text{kg K}} \quad \text{Heat capacity of air}$$

$$\alpha := \frac{c_p \cdot \rho}{3600} = 0,335 \frac{\text{Pa}}{\text{K}} \quad \text{This approximation is commonly used in HVAC calculations to estimate the sensible heat removed through ventilation.} \quad (3.2)$$

$$Q_{\text{cool}} := \alpha \cdot (V_{\text{vent}} + V_{\text{fans}}) \cdot \Delta T \quad \text{Sensible cooling load due to airflow} \quad (3.3)$$

$$Q_{\text{cool}} = 0,2557 \text{ W} \quad \text{Removing } \approx 0.25 \text{ joules of heat per second}$$

4. Lighting:

$$PPFD_{required} := 400 \frac{\text{micron}}{\text{m}} \frac{\text{mol}}{\text{s}} \quad \text{to supply a range of needs of vegative growth from seedlings to leafy greens}$$

$$PPFD_{sunlight} := 200 \frac{\text{micron}}{\text{m}} \frac{\text{mol}}{\text{s}} \quad \begin{array}{l} \text{light intensity of sunlight averaged over 12 hour period} \\ \text{asuming sunlight averages } 200\mu\text{mol/m}^2/\text{s} \end{array}$$

$$PPFD_{LED} := PPFD_{required} - PPFD_{sunlight} = 200 \frac{\text{micron}}{\text{m}} \frac{\text{mol}}{\text{s}} \quad \text{PPFD needed to match required level}$$

$$Grow_Area := 850 \text{ mm} \cdot 850 \text{ mm} \quad \text{estimated maximum plant coverage within greenhouse}$$

$$Light_Efficiency := 1,8 \frac{\text{micron}}{\text{J}} \quad \begin{array}{l} \text{micromols of PAR produced per Joule of electricity consumed} \\ \text{asuming efficiency of LED to be } 1,8 \mu\text{mol/J} \end{array}$$

$$Power_{LED} := PPFD_{LED} \cdot \frac{Grow_Area}{Light_Efficiency} = 80,2778 \text{ W} \quad \text{LED power required to supplement lighting} \quad (4)$$

5. Perspex Wind Loading:

$$t := 3 \text{ mm} \quad \text{Plate thickness} \quad \sigma_m := 72 \text{ MPa} \quad \text{Tensile strength of Plexiglass from datasheet sourced through Maizey Plastics}$$

$$\nu := 0,35 \quad \text{Poisons ratio} \quad b = 0,85 \text{ m}$$

$$E_{perspex} := 3,2 \text{ GPa}$$

$$v_{wind} := 40 \frac{\text{km}}{\text{hr}} \quad \text{Maximum gust winds asumed to be up to 40km/h and acting normal to panel}$$

$$D := \frac{t^3 \cdot E_{perspex}}{12 \cdot (1 - \nu^2)} = 8,2051 \text{ N m} \quad \begin{array}{l} \text{Flexural Rigidity of the Perspex plate asuming that it is} \\ \text{uniformly loaded and simply supported at all ends} \end{array} \quad (5.1)$$

$$P := 0,5 \cdot \rho \cdot v_{wind}^2 = 74,0741 \text{ Pa} \quad \text{Dynamic pressure} \quad (5.2)$$

$$y_{pmax} := P \cdot \frac{b^4}{D \cdot 64} = 0,0736 \text{ m} \quad \text{Maximum deflection in a uniformly loaded simply supported rectangular plate using Kirchhoff-Love and Mindlin-Reissner Plate theory} \quad (5.3)$$

$$\sigma_{pmax} := \frac{6 \cdot P \cdot b^2}{t^2} = 35,679 \text{ MPa} \quad \text{Maximum bending stress in a uniformly loaded simply supported rectangular plate using Kirchhoff-Love and Mindlin-Reissner Plate theory} \quad (5.4)$$

Tensile strength of perspex $\approx 70 \text{ MPa}$

$$S_{perspex} := \frac{\sigma_m}{\sigma_{pmax}} = 2,018 \quad \text{Factor of safety for perspex due to optimistic wind loading at 40km/h} \quad (5.5)$$

6. Aluminium Frame Structure:

$$I := 4,92 \cdot 10^{-9} \text{ m}^4$$

Moment of inertia

$$c := 10 \text{ mm}$$

Distance from centroid to end of aluminium cross section

$$E := 69 \text{ GPa}$$

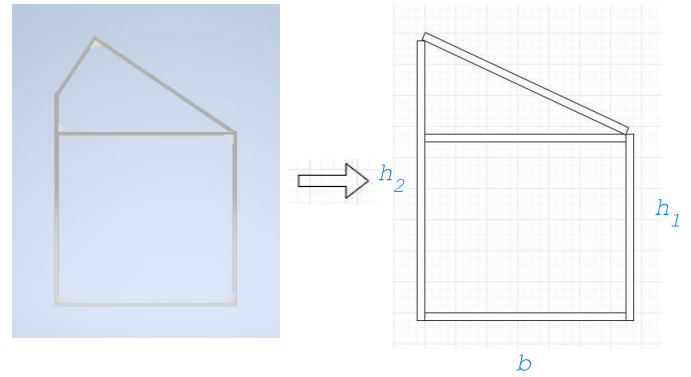
Modulus of elasticity from Macsteel

$$\sigma_y := 310 \text{ MPa}$$

Aluminium yield strength

$$C_d := 1,2$$

Drag coefficient



$$A_{side} := b \cdot h_1 + b \cdot \frac{h_2 - h_1}{2} \quad \text{Area of side panel}$$

$$A_{side} = 0,7735 \text{ m}^2$$

$$A_{section} := 80 \text{ mm}^2$$

Area of aluminium square tubing section

$$v_{wind} := 40 \frac{\text{km}}{\text{hr}}$$

Maximum gust winds assumed to be up to 40km/h

$$\rho_{aluminium} := 2,7 \frac{\text{g}}{\text{m}^3}$$

Density of aluminium from Macsteel

$$P := 0,5 \cdot \rho \cdot v_{wind}^2 \cdot C_d = 88,8889 \text{ Pa} \quad \text{Pressure on panels due to wind loading (5.2)}$$

$$F := P \cdot A_{side} = 68,7556 \text{ N}$$

Total force on side panel

$$w_{wind} := \frac{F}{2 \cdot h_1} = 41,9241 \frac{\text{N}}{\text{m}}$$

Force per meter of height due to wind

For aluminium section A-B assume that this member resists half of the force distributed across the entire side panel as a uniform load

Considering the vertical bar to be simply supported and resist uniform load. The max moment occurs at the midpoint of the bar

$$M_{wind} := w_{wind} \cdot \frac{h_1^2}{8} = 3,5237 \text{ N m}$$

Simple supports and uniform load from Shigleys Mechanical Design 10th edition

(6.1)

$$\sigma_{wind} := M_{wind} \cdot \frac{c}{I} = 7,162 \text{ MPa}$$

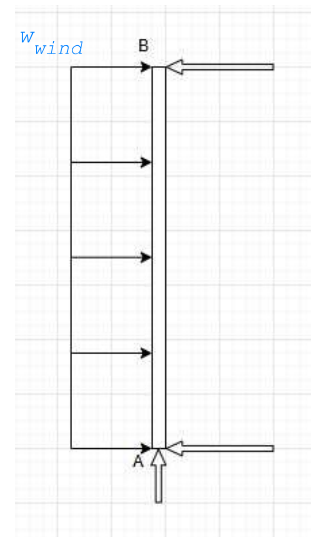
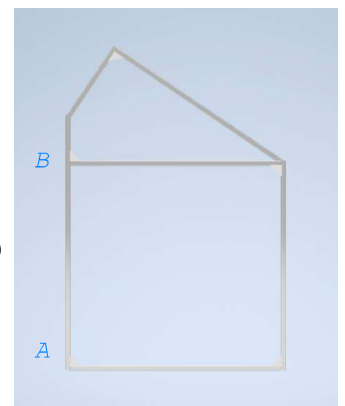
Max stress due to uniform wind load

(6.2)

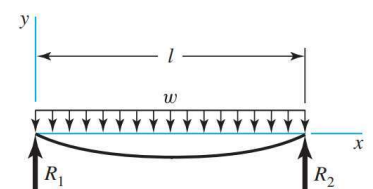
$$y_{max} := \frac{-5 \cdot w_{wind} \cdot h_1^4}{384 \cdot E \cdot I} = -0,0007 \text{ m}$$

Maximum deflection at midpoint of bar

(6.3)



7 Simple supports—uniform load



$$m_{frame} := 0,3 \frac{\text{kg}}{\text{m}} \cdot \left(2 \cdot \left(h_1 + h_2 + b + \sqrt{b^2 + (h_1 - h_2)^2} \right) + 5 \cdot l \right) = 3,6233 \text{ kg} \quad \text{Mass of all aluminium sections}$$

$$A := l \cdot h_1 \cdot 2 + l \cdot h_2 + \left(2 \cdot b \cdot \frac{(h_2 - h_1)}{2} + 2 \cdot b \cdot h_1 \right) \quad \text{Area of all sides of perspex}$$

$$V := A \cdot t = 0,0126 \text{ m}^3 \quad \text{Volume of all perspex}$$

$$m_{panels} := V \cdot \rho_{perspex} = 14,9476 \text{ kg}$$

$$m_{total} := m_{frame} + m_{panels} = 18,5709 \text{ kg} \quad \text{Total mass of structure} \quad (6.3)$$

$$F_{weight} := \frac{m_{total}}{4} \cdot 9,81 \frac{\text{m}}{\text{s}^2} = 45,5451 \text{ N} \quad \text{Assuming each vertical aluminium member supports a fourth of the total weight.} \quad (6.4)$$

$$\sigma_{weight} := \frac{F_{weight}}{A_{section}} = 0,5693 \text{ MPa} \quad \text{Stress due to weight of structure} \quad (6.5)$$

$$\sigma_{total} := \sigma_{wind} + \sigma_{weight} = 7,7314 \text{ MPa} \quad \text{Yield strength of aluminium: 100–250 MPa} \rightarrow \text{well within limits}$$

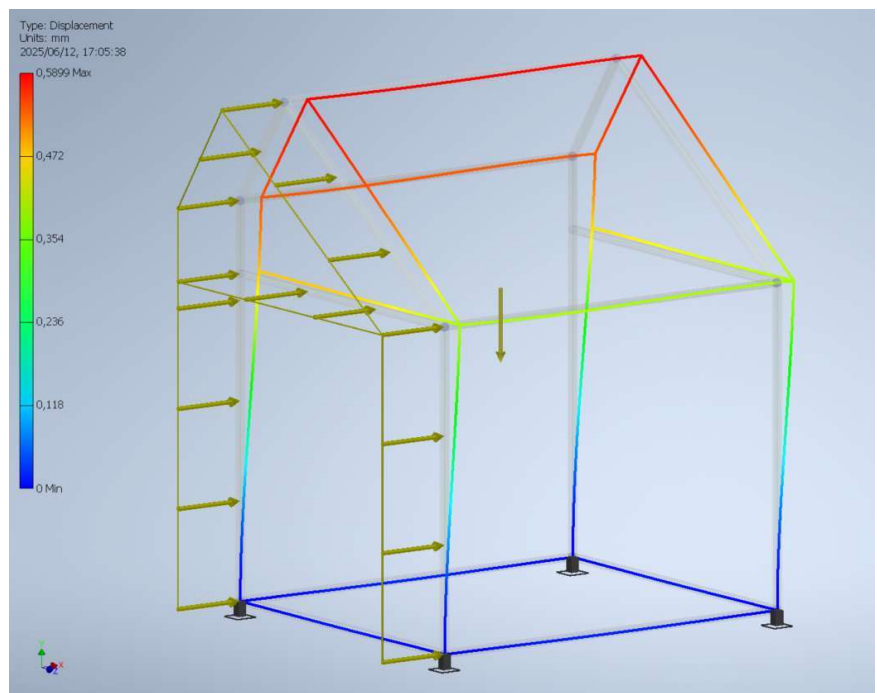
$$S_{aluminium} := \frac{\sigma_y}{\sigma_{total}} = 40,0965 \quad \text{Factor of safety for horizontal aluminium sections under wind loading}$$

6.1 Frame Analysis Using Inventor Environment:

$$Load_{wind} := 0,0038 \text{ N m}$$

For continuous load of 18N over all 4739mm of tubing on the side profile due to a 20km/h wind acting perpendicular to the panel. assuming 4 fixed points at the base of the greenhouse. assuming the force is resisted equally by each member on the side of the greenhouse for use in frame analysis simulation.

Maximum displacement is located at the apex of the greenhouse is limited to only 0.51mm



7.1 Fan Sizing:

$$V := b \cdot h_2 \cdot l - \frac{(h_2 - h_1)}{2} \cdot b \cdot l = 0,7735 \text{ m}^3$$

Total volume of greenhouse

$$flow := \frac{V}{\text{min}} = 0,7735 \frac{\text{m}^3}{\text{min}}$$

Total flow required by fans using standard airflow sizing of one entire volume exchange per minute (7.1)

$$CFM := flow \cdot 35 \frac{\text{ft}^3}{\text{m}^3} = 27,0725 \frac{\text{ft}^3}{\text{min}}$$

Total flow in CFM

7.2 HAF Requirements:

$$A_{\text{floor}} := b \cdot l = 0,85 \text{ m}^2$$

Total usable area of greenhouse

$$A_{\text{floorft}} := A_{\text{floor}} \cdot 10,764 \frac{\text{ft}^2}{\text{m}^2} = 9,1494 \text{ ft}^2$$

$$CFM_{\text{HAF}} := A_{\text{floorft}} \cdot 2 \frac{\text{ft}^3}{\text{min}} = 18,2988 \frac{\text{ft}^3}{\text{min}}$$

CFM Required using standard HAF sizing of 2CFM fan capacity per square foot of greenhpouse area (7.2)

8. Power Supply:

8.1 Battery sizing:

Components	Power	Usage 24 hr	Wh 24 hr	
Heater	120 W	2 hr	240 Whr	~2 hours (nighttime)
Grow – Light	100 W	3 hr	300 Whr	~3 hours (morning/afternoon)
Ventilation – Fans	10 W	8 hr	80 Whr	~8 hours (daytime)
Sensors – Actuators – Controller	10 W	24 hr	240 Whr	~24 hours (intermittent use)
TOTAL	■	■	860 Whr	

$$Draw := 860 \text{ W hr}$$

$$Voltage := 12 \text{ V}$$

$$DoD := 80 \%$$

Allowable Depth of Discharge

$$Capacity := \frac{Draw}{Voltage \cdot DoD} = 89,5833 \text{ A hr}$$

Minimum required battery capacity (8.1)

8.2 Solar Panel Sizing:

$$Battery_{Ah} := 100 \text{ A hr}$$

Selecting a 100Ah battery

$$hours_{\text{day}} := 6 \text{ hr}$$

Asuming 6 hours of sunlight per day

$$W_{\text{panel}} := Battery_{Ah} \cdot \frac{Voltage}{hours_{\text{day}} \cdot 0,8} = 250 \text{ W}$$

Asuming effeciency of 80% (8.2)

A solar array rated to provide atleast 250W would be able to charge the battery fully in one day asuming a moderate system efficiency of 80%